Demonstration of Proposed Features

Date	2 July 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S. No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Emotion Classification Engine (BiLSTM & BERT)	Predicts the learner's emotional state — Bored, Confident, Confused, Curious, or Frustrated — from free-text input	Implemented	Yes	Both models classify emotions reliably
2	Rule-Based Keyword Enhancement	Refines model predictions using keyword-based cues for nuanced cases	Implemented	Yes	Improves accuracy on edge cases
3	Mixed-Emotion Detection	Detects blended emotional states (e.g., Curious + Confused) rather than a single label	Implemented	Yes	Provides richer emotional insight
4	Gemini-Generated Guidance	Generates personalized tips, next steps, and encouragement based on the detected emotion	Implemented	Yes	Responses feel supportive and relevant
5	Dual-Model Comparison	Displays BiLSTM vs BERT predictions side-by-side for transparency	Implemented	Yes	Improves reliability and trust in results
6	Interaction Logging	Logs each interaction to a CSV file to support continuous learning	Implemented	Yes	History builds up correctly over sessions
7	Analytics Dashboard	Visualizes emotion trends across sessions through charts	Implemented	Yes	Charts render as expected
8	Streamlit App Interface	Lets learners enter their problem context and toggle AI responses in a simple UI	Implemented	Yes	Fully functional frontend

Feature Implementation Summary:

Particular	Count
Total Features Proposed	8
Total Features Implemented	8
Total Features Demonstrated	8
Overall Implementation Rate (%)	100%